Angel Barrera

Data & AI Governance | Data Science | Information Systems

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RESEARCH PROFILE

My research interests focus on how Data Governance, AI Governance, organizational capabilities, and technical foundations influence whether AI initiatives move from experimentation into sustained operational use. My background combines applied data science research with enterprise experience in data governance, metadata, data quality, analytics, and information systems. I am particularly interested in organizational AI readiness, responsible implementation, governance maturity, and the conditions that allow organizations to scale AI while maintaining accountability and measurable value.

RESEARCH INTERESTS

Data Governance | AI Governance | Responsible AI | Organizational AI Readiness | AI Adoption and Scaling | Information Systems | Enterprise Architecture | Governance Maturity | Digital Transformation

EDUCATION

Master of Science in Applied Data Science — **Aug 2024 – Dec 2025**
East Tennessee State University — *Johnson City, Tennessee, USA*

- GPA: **3.79/4.00**; 33 graduate semester credits.
- Relevant coursework: Artificial Intelligence, Machine Learning, Data Management, Predictive Analytics, Statistical Methods I–II, Data Analytics & Visualization, Decision Support Systems, and Advanced Data Analysis.
- Completed the program's Master's Culminating Experience through applied data science research associated with the Oak Ridge National Laboratory collaboration described below.

Computer Systems Engineering — **Jan 2019 – May 2024**
Universidad Evangélica de El Salvador — *San Salvador, El Salvador*

- Final GPA: **8.94/10.00**.
- Relevant coursework: Algorithms, Programming, Databases, Software Engineering, Systems Analysis, Data Structures, Statistics and Probability, Research Methods I–II, Management Information Systems, Information Management and Technology, Business Intelligence, and Systems Audit.
- Completed formal graduation research on the implementation of Business Intelligence and data analytics at Fundación Pro-Educación de El Salvador (FUNPRES).

RESEARCH EXPERIENCE

Principal Investigator, Data & AI Governance Research Proposal — **2026 – Present**
Universidad Evangélica de El Salvador — *San Salvador, El Salvador*

From AI Adoption to AI Success: Data and AI Governance as Predictors of Successful AI Implementation

- Developed a research framework to examine whether the maturity of Data Governance and AI Governance can help predict whether organizational AI initiatives reach production, meet their objectives, remain operational, and generate measurable value.
- Designed a prospective longitudinal study at the AI-initiative level, incorporating governance maturity, organizational characteristics, strategy, leadership, digital maturity, technical capability, investment, and implementation complexity.
- Proposed the development and validation of a governance-maturity instrument informed by academic literature, NIST AI RMF, and ISO/IEC 42001.
- Defined a quantitative analysis approach using multivariable logistic regression, internal validation, calibration and discrimination assessment, and exploratory analysis of maturity thresholds and differences across organizational contexts.
- Grounded the study in the Resource-Based View, dynamic capabilities theory, IT business-value research, and the Data and AI Governance literature.

Applied Researcher / Data Science Intern **Aug 2025 – Dec 2025**
Oak Ridge National Laboratory Collaboration *Program Project / ETSU*

3D Deep Learning for Plant Organ Segmentation using Dynamic Edge Convolutional Neural Networks

Research code and documentation: github.com/angomezu/geometric-deep-learning-plant-organ-segmentation

- Conducted applied research on semantic segmentation of LiDAR-based 3D point clouds for plant phenotyping in collaboration with Oak Ridge National Laboratory.
- Co-developed a structured annotation procedure for classifying point-cloud data into stem, leaf, support stake, and background classes and prepared labeled data for supervised model training.
- Designed and evaluated a Dynamic Edge Convolutional Neural Network using PyTorch and PyTorch Geometric, with geometric feature engineering, class-aware loss functions, bootstrap evaluation, and post-processing.
- Achieved **65.58% mean Intersection over Union** and approximately **82% stem recall** while working with substantial class imbalance and geometric ambiguity.
- Contributed to the research codebase, experimental evaluation, technical documentation, and final presentation.

Undergraduate Graduation Research **2024**
Universidad Evangélica de El Salvador / FUNPRES *San Salvador, El Salvador*

Implementation of Power BI for Monitoring, Evaluation, and Data Analysis in the "Aprendo para Brillar" Project

Original title: Implementación de Power BI para el Monitoreo, Evaluación y Análisis de Datos en la Ejecución de los Circuitos del Proyecto "Aprendo para Brillar"

Institutional repository: 138.99.0.237/handle/123456789/769

- Co-authored the formal graduation monograph required for completion of the Computer Systems Engineering degree.
- Studied how Business Intelligence and improved access to organizational data affected monitoring, operational management, and decision-making within an education-focused nonprofit program.
- Designed a centralized analytical data model and Power BI reporting environment, automated reporting processes, developed operational dashboards, and supported implementation and user training.
- Evaluated results related to information availability, operational errors, monitoring capability, reporting efficiency, and project decision-making.

PROFESSIONAL EXPERIENCE

Data Governance Lead **Mar 2026 – Present**
CenterPoint Energy *Houston, Texas, USA*

- Lead enterprise Data Governance across **20 business domains and 27,428 in-scope assets**, working with Domain Owners, Data Stewards, Technical Stewards, engineers, analysts, data scientists, and leadership on ownership, standards, controls, stewardship, escalation, and governance reporting.
- Helped advance governance execution from **0 to 813 governed assets**, with all 20 business domains establishing their first governed assets and associated stewardship practices.
- Built automated reconciliation across **26,980 assets and 935 datasets**, identifying **6,573 metadata and monitoring gaps** and establishing a repeatable process for investigation and remediation.
- Designed controls for sensitive and critical data, identifying **67 PI/CEII columns** omitted from curated reporting and developing workload-informed approaches for prioritizing Critical Data Assets and Critical Data Elements.
- Designed the Data Governance Platform and Data Governance Intelligence Foundation to support metadata, Data Quality, governance workflows, AI Governance, institutional knowledge, and evidence-based review.
- Use metadata, lineage, ontology, business context, and AI-assisted workflows to support governance analysis, validation, troubleshooting, and decision-making.

Data Analytics Engineer **Jan 2023 – Aug 2024**
AES El Salvador *San Salvador, El Salvador*

- Led data and analytics modernization across finance, engineering, and operations in a regulated energy environment, developing a governed BigQuery semantic and metric layer supporting more than **183 Power BI reports**.
- Developed SAP-to-BigQuery reconciliation and validation controls that improved data quality, consistency, traceability, and auditability while reducing reporting latency by more than **40%**.
- Established metadata, data-dictionary, lineage, documentation, and analytics release-governance practices using Atlan and Azure DevOps.

Graduate Assistant **Aug 2024 – Dec 2025**
East Tennessee State University, International Enrollment and Services *Johnson City, Tennessee, USA*

- Developed an end-to-end Power BI report analyzing international students' sense of belonging and supported stakeholder reporting, student engagement, onboarding, and resource-navigation initiatives.

Data Engineer, EXECUTEL Nov 2021 – Dec 2022
Data Quality, Python and SQL data processing, operational analytics, CRM/ERP implementation, technical requirements, and reporting automation.

DevOps Engineer, Resultier Apr 2022 – Nov 2022
AWS EC2, Linux, Docker, GitHub Actions, deployment automation, and CI/CD.

Operations Data Analyst, The Office Gurus Jul 2019 – Oct 2021
Operational data analysis, financial and client data integrity, Excel/VBA automation, reporting, and database-supported analytics.

RESEARCH METHODS & TECHNICAL SKILLS

Research Methods: Quantitative Research Design | Longitudinal Studies | Predictive Modeling | Regression | Internal Validation | Bootstrap Evaluation | Calibration and Discrimination Assessment | Research Instrument Design | Literature Review

Data Science and Governance: Python | SQL | PyTorch | PyTorch Geometric | Machine Learning | Deep Learning | Data Governance | AI Governance | Data Quality | Metadata | Lineage | Data Modeling | Business Intelligence

Platforms and Engineering: GCP | BigQuery | Vertex AI | Cloud Run | Atlan | Monte Carlo | Apache Iceberg | Apache Polaris | Power BI | Tableau | SAP S/4HANA | REST / GraphQL APIs | Git | GitHub | Docker | Linux

RECOGNITIONS

Atlan Context Agents Accelerator, Context Pioneer 2026
Recognized for applied enterprise work involving metadata, lineage, business context, ontology, explainability, AI-assisted governance workflows, and human review.

LANGUAGES

- Spanish: Native
- English: Professional and Academic Proficiency